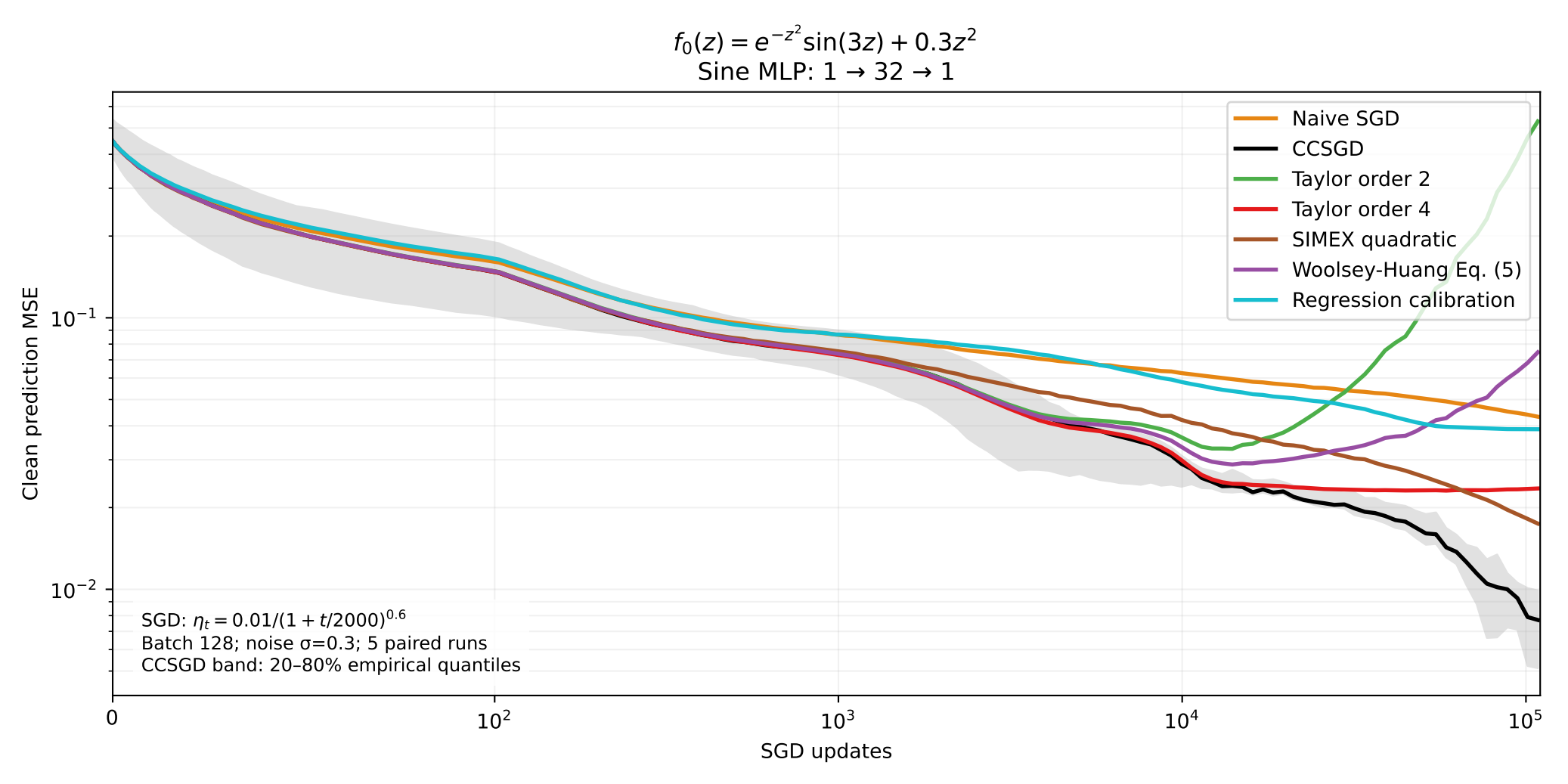
$f_0(z) = e^{-z^2}\sin(3z) + 0.3z^2$
Sine MLP: 1 → 32 → 1
Naive SGD
CCSGD
Taylor order 2
Taylor order 4
SIMEX quadratic
Woolsey-Huang Eq. (5)
Regression calibration
Clean prediction MSE
10^{-1}
10^{-2}
SGD: $\eta_t = 0.01/(1 + t/2000)^{0.6}$
Batch 128; noise σ=0.3; 5 paired runs
CCSGD band: 20–80% empirical quantiles
0
10^2
10^3
10^4
10^5
SGD updates